

PRESENTABLE QUASICATEGORIES AS MODELS OF LIMIT SKETCHES

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ABSTRACT. The presentable ∞ -categories have a key role in the study of the higher order topos, as can be seen in the works of Lurie. On the other hand, the theory of sketches by Ehresmann is a natural framework to study homotopy coherent structures. In the context of ordinary category theory, there exists a characterization of the locally presentable categories by sketches: the representation theorem of Adámek-Rosický. Our goal is to study whether a similar characterization in the context of ∞ -categories makes sense.

Representation Theorem (Adamek and Rosicky [1]). *The following are equivalent:*

- (i) *Presentable categories.*
- (ii) *Localizations of categories of presheaves over some small category.*
- (iii) *Categories of models of limit sketches.*

There is a similar representation theorem in the context of ∞ -categories?

The rest of the notes are divided in two sections. First, we will review the basic definitions of sketches and presentable categories, and state the representation theorem in the context of ordinary categories. In the second section, we repeat this process but in the context of ∞ -categories. During these notes, we denote ∞ -category to mean quasicategory, see the works of Lurie [4] for more details about this model of higher categories.

1. REPRESENTATION THEOREM ON CATEGORIES

Let K be a small category, and C be a category. Recall that a diagram is a functor with a small category as domain. Given any object $x \in C$, define $\text{const}^x : K \rightarrow C$ as the constant diagram on x , i.e., for all object $k \in \text{Obj } K$ and all morphism $f \in \text{Mor } K$

$$\text{const}^x(k) = x, \quad \text{and} \quad \text{const}^x(f) = \text{id}_x.$$

A *cone* (K, D, x, α) is a quadruple of a small category K of indices, a diagram $D : K \rightarrow C$, an object $x \in \text{Obj } C$, and a natural transformation $\alpha : \text{const}^x \Rightarrow D$. A cone (K, D, x, α) is called a *limit cone* if, for all $y \in \text{Obj } C$, the post-composition of α is an isomorphism

$$\text{Hom}_C(y, x) \xrightarrow{\cong} \text{Cones}(K, D, y) := \text{Hom}_{\text{Fun}(K, C)}(\text{const}^y, D),$$

where $\text{Cones}(K, D, y)$ is the set of all cones (K, D, y, β) for some $\beta : \text{const}^y \Rightarrow D$.

Definition 1.1 (Bastiani and Ehresmann [2]). Let A be a small category. A *limit sketch* $S = (A, L)$ is a pair of a small category A , and a set of cones $L = \{(K_i, D_i, x_i, \alpha_i)\}_{i \in I}$. A *model* of a limit sketch $S = (A, L)$ is a functor $M : A \rightarrow \mathbf{Set}$ that sends the cones of L to limit cones in $\mathbf{Set}$, i.e., that for every cone $(K_i, D_i, x_i, \alpha_i) \in L$, the map induced by the limit in $\mathbf{Set}$ is an isomorphism:

$$t_i : M(x_i) \xrightarrow{\cong} \lim_{K_i} M \circ D_i$$

The category of all models of a sketch S is denoted $\text{Mod}(S)$.

Example 1.2 (Lawvere theories). A Lawvere theory T is a category with finite products in which every object is isomorphic to a power of a selected object $x \in \text{Obj } T$. The category of algebras Alg_T over a Lawvere theory T is the category of models of a limit sketch $S = (T, L)$, where L is the set of all product cones. Many categories are examples of categories of algebras over a Lawvere theory, like the category of groups, R -modules, and Lie algebras.

Recall that a category is *accessible* if there exists a regular cardinal κ such that the category admits κ -filtered colimits, and there is a set of κ -compact objects that generate the category under κ -filtered colimits. Then, a category is *presentable* if it is accessible and admits small limits. Presentable categories first appeared in the work of Gabriel and Ulmer [3], and similar results to the representation theorem were discussed in the works of Makkai and Paré [5].

Theorem 1.3 (Representation theorem, Adamek and Rosicky [1]). *The following are equivalent:*

- (i) C is a presentable category.
- (ii) C is equivalent to a localization of the category of presheaves over some small category.
- (iii) C is equivalent to the category of models $\text{Mod}(S)$ of a limit sketch S .

2. REPRESENTATION THEOREM ON INFINITY CATEGORIES

Let $\mathcal{K}$ be a small ∞ -category, and $\mathcal{C}$ be an ∞ -category. Recall that a diagram is a functor with a small ∞ -category as domain. Given any object $x \in \text{Obj } \mathcal{C}$, define $\text{const}^x : \mathcal{K} \rightarrow \mathcal{C}$ as the constant diagram on x , i.e., for all object $k \in \mathcal{K}_0$ and all n -morphism $f \in \mathcal{K}_n$ with $n \geq 1$

$$\text{const}_0^x(k) = x, \quad \text{and} \quad \text{const}_n^x(f) = (s_0)^n(x).$$

A *cone* $(\mathcal{K}, D, x, \alpha)$ is a quadruple of a small ∞ -category $\mathcal{K}$ of indices, a diagram $D : \mathcal{K} \rightarrow \mathcal{C}$, an object $x \in \text{Obj } \mathcal{C}$, and a natural transformation $\alpha : \text{const}^x \Rightarrow D$. A cone $(\mathcal{K}, D, x, \alpha)$ is called a *limit cone* if, for all $y \in \text{Obj } \mathcal{C}$, the post-composition of α is a weak equivalence of Kan complexes

$$\text{Map}_{\mathcal{C}}(y, x) \xrightarrow{\cong} \text{Cones}(\mathcal{K}, D, y) := \text{Map}_{\text{Fun}(\mathcal{K}, \mathcal{C})}(\text{const}^y, D),$$

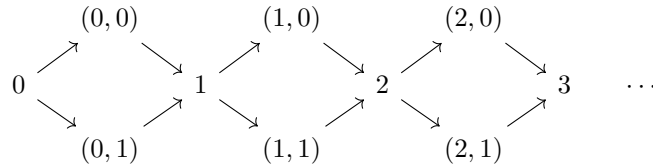
where $\text{Cones}(\mathcal{K}, D, y)$ is the set of all cones $(\mathcal{K}, D, y, \beta)$ for some $\beta : \text{const}^y \Rightarrow D$.

Definition 2.1. Let A be a small category. A *limit sketch* $S = (\mathcal{A}, L)$ is a pair of a small ∞ -category $\mathcal{K}$, and a set of cones $L = \{(\mathcal{K}_i, D_i, x_i, \alpha_i)\}_{i \in I}$. A *model* of a limit sketch $S = (\mathcal{A}, L)$ is a functor $M : \mathcal{A} \rightarrow \mathbf{Spaces}$ that sends the cones of L to limit cones in $\mathbf{Spaces}$, i.e., that for every cone $(\mathcal{K}_i, D_i, x_i, \alpha_i) \in L$, the map induced by the limit in $\mathbf{Spaces}$ is a weak equivalence of Kan complexes:

$$t_i : M(x_i) \xrightarrow{\sim} \lim_{\mathcal{K}_i} M \circ D_i$$

The category of all models of a sketch S is denoted $\text{Mod}(S)$.

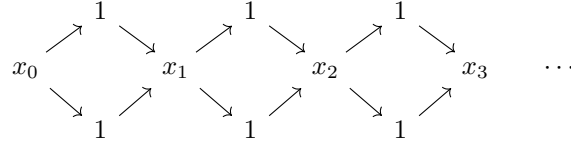
Example 2.2 (Spectra). Let A be the small category with objects $\mathbb{N} \sqcup (\mathbb{N} \times \{0, 1\})$, and generating morphisms $f_{i,j} : i \rightarrow (i, j)$ and $g_{i,j} : (i, j) \rightarrow (i+1, j)$ for every $i \in \mathbb{N}$ and $j \in \{0, 1\}$, i.e., the category of the following shape:



In addition, define the set

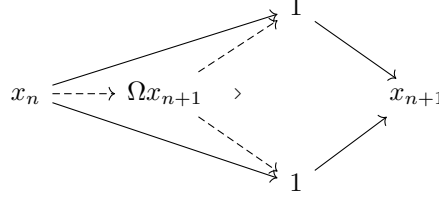
$$L = \{(\emptyset, D_{i,j} : \emptyset \rightarrow A, (i, j), \alpha_{i,j} : \text{const}^{(i,j)} \Rightarrow D_{i,j})\}_{i \in \mathbb{N}, j \in \{0,1\}}.$$

Then, a model $M : A \rightarrow \mathbf{Spaces}$ for the sketch $S = (A, L)$ is a diagram



where each (i, j) is replaced by the terminal object, and a sequence of objects $x_n \in \mathbf{Spaces}$ is selected.

A model of this sketch is the same as having pointed objects $1 \rightarrow x_n$ for every $n \in \mathbb{N}$, and maps $x_n \rightarrow \Omega x_{n+1}$ (by the universal property of the pullback):



Hence, each model of S is by definition a *pre-spectrum*, and $\text{Mod}(S)$ is the ∞ -category of pre-spectra.

If we want spectra, we need to add more cones to S . Define I as the index category with objects $\{0, 1, 2\}$ and generating morphisms $f : 0 \rightarrow 1$ and $g : 2 \rightarrow 1$. Then, take the set of cones

$$L' = L \sqcup \{(I, D_n, n+1, \beta_n) \mid D_n : I \rightarrow A, \beta_n : \text{const}^{n+1} \Rightarrow D_n\}_{n \in \mathbb{N}}.$$

Hence, a model for S is a pre-spectrum such that

$$x_n \simeq \text{pullback of } \{1 \rightarrow x_{n+1} \leftarrow 1\} := \Omega x_{n+1}.$$

In other words, a model for S is a spectrum, and $\text{Mod}(S)$ is the ∞ -category of spectra.

Recall that an ∞ -category is *accessible* if there exists a regular cardinal κ such that the ∞ -category is locally small, it admits κ -filtered colimits, and there is some essentially small sub- ∞ -category of κ -compact objects that generate the ∞ -category under κ -filtered colimits. Then, a category is *presentable* if it is accessible and admits small limits.

Theorem 2.3 (Representation theorem I, Simpson [6] and Lurie [4]). *An ∞ -category $\mathcal{C}$ is presentable if, and only if, $\mathcal{C}$ is equivalent to an accessible localization of the ∞ -category of presheaves over some small ∞ -category.*

Goal 2.4 (Representation theorem II). *An ∞ -category $\mathcal{C}$ is presentable if, and only if, $\mathcal{C}$ is equivalent to the category of models $\text{Mod}(S)$ for a limit sketch S .*

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